Non-Markovian Gravitational Decoherence: A Gaussian-Onset Alternative to the Diósi–Penrose Exponential

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(Dated: Version of 11 June 2026 — preprint, not peer reviewed)

Gravitationally induced decoherence, in the Diósi–Penrose (DP) form, predicts that a spatial superposition of a massive body loses coherence at a rate set by the gravitational self-energy E_G of the mass-density difference between the branches, with characteristic time $\tau_G = \hbar/E_G$. The DP law is exponential in time, a form that follows from an implicit Markov (memoryless, divisible) assumption about the underlying stochastic dynamics. We examine gravitational decoherence within the framework of *indivisible stochastic processes* (ISP) of Barandes, whose defining feature is the failure of the divisibility (Chapman–Kolmogorov) property. Indivisibility motivates treating the gravitational record-mismatch noise as non-Markovian (finite memory time τ_c) rather than white. ISP motivates this finite-memory replacement, and an explicit division-event model (Sec. IV C) grounds the resulting exponential kernel and pins τ_c—modulo one stated ansatz, with the energy scale inherited from DP. We show that any such finite-memory noise—a finite, regular covariance with no white component—generically replaces the DP exponential with a non-exponential coherence decay whose short-time exponent is quadratic: Gaussian, $C(T) = C(0)\exp[-\frac{1}{2}(T/\tau_G)^2]$, when the mismatch-energy fluctuation scale is of order E_G and τ_c is at least of order τ_G, crossing over to a DP-slope exponential tail at later times. The strict DP exponential is recovered in the white-noise limit ($\tau_c \to 0$ at fixed noise power), where the Gaussian onset window closes. The distinction is experimentally sharp and *rate-robust* (it does not rely on fitting a single exponential rate): the onset shape of the residual gravitational log-coherence is linear in T for Markovian collapse and quadratic (zero initial slope, a Zeno-like plateau) for the indivisible case. We give threshold masses and separations for levitated-nanosphere, matter-wave, and mechanical-cat-state platforms, set out differential strategies (notably a density lever and common-mode subtraction) to separate the gravitational onset from environmental backgrounds, and confront the radiation and heating bounds, which constrain only the model's regularization length—the per-nucleon emission ($\propto 1/R_0^3$), decoupled from the (R_0-independent) collective decoherence the interferometer measures ($\propto GM^2/R$)—so the finite-memory modification changes the *onset shape*, not the radiation phenomenology. We are explicit about scope: the quadratic onset is the generic signature of non-Markovian decoherence and is shared with other coloured-noise collapse models, so the test falsifies *strict* Markovian DP/CSL and would be evidence for a non-Markovian fundamental channel consistent with ISP, but does not by itself single out ISP.

I. INTRODUCTION

The persistence of quantum coherence for increasingly massive systems is one of the sharpest open questions at the quantum–classical and quantum–gravity interfaces. Objective-collapse models posit a fundamental, observer-independent mechanism that suppresses macroscopic superpositions [4–7]. Among them, the proposals of Diósi [2, 3] and Penrose [1] tie this suppression specifically to gravity: a superposition of two distinct mass distributions is a superposition of two distinct spacetime geometries, and the gravitational self-energy of their difference sets a decoherence time $\tau_G = \hbar/E_G$. This "Diósi–Penrose" (DP) decoherence is now an active experimental target, both directly through matter-wave and optomechanical interferometry [15, 18, 22, 25, 26] and indirectly through spontaneous-radiation searches [8, 9], the latter having already excluded the simplest, parameter-free DP model [8].

The DP coherence law is exponential, $C(T) = C(0)\exp(-T/\tau_G)$. This functional form is not an independent prediction of gravity; it is a consequence of treating the underlying collapse noise as *white* (delta-correlated in time), equivalently of assuming the reduced dynamics forms a Markovian semigroup. White noise is an idealization [10], and the open-systems literature shows that non-Markovian (coloured) environments generically produce non-exponential coherence decay, with a universal quadratic (Zeno-like) onset at short times [14, 20, 21].

In this paper we ask what gravitational decoherence looks like when the underlying dynamics is taken to be *indivisible* in the sense of Barandes' stochastic-quantum correspondence [11, 12]. An indivisible stochastic process is one whose transition law does *not* factorize through intermediate times,

$$\Gamma(t_2, t_0) \neq \Gamma(t_2, t_1)\,\Gamma(t_1, t_0), \tag{1}$$

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i.e. it violates exactly the divisibility property that underlies the Markovian semigroup. Our central result is that this feature—finite memory in the gravitational channel—replaces the DP exponential with a non-exponential law whose short-time exponent is quadratic (Gaussian when the fluctuation scale is of order E_G), and that the difference is experimentally accessible through a *shape-based* onset measurement that needs no fitted rate. We recover the DP exponential in the white-noise (Markovian) limit, identify the new time scale τ_c that controls the crossover, and confront the existing spontaneous-radiation bound.

Our claims are deliberately bounded. Indivisibility *motivates* a finite-memory gravitational channel; an explicit division-event model (Sec. IV C) then grounds the exponential kernel and pins τ_c, at the cost of one stated ansatz, while the amplitude (energy scale) is inherited from DP. The mechanism is a *reconstruction* of gravitational decoherence within a particular stochastic ontology, and the new, falsifiable content is the non-exponential onset, not the energy scale. The quadratic onset is the generic signature of non-Markovianity and is not unique to ISP. The robust, central claim is that the DP exponential is a Markovian idealization and that finite memory yields a falsifiable onset-shape discriminator; the radiation and heating bounds (Sec. VIII) constrain only the model's regularization length and decouple from the (R_0-independent) decoherence, so they do not threaten the prediction. We make these limitations explicit in Sec. IX.

II. INDIVISIBLE STOCHASTIC DYNAMICS AND THE RECORD-STABILIZATION HYPOTHESIS

Barandes' stochastic-quantum correspondence [11, 12] establishes that a broad class of quantum systems can be represented as stochastic processes over a configuration space in which the configuration is a definite quantity ("be-able") at all times, with the unitary, Hilbert-space description emerging as an exact reformulation. The processes that correspond to generic quantum evolution are *indivisible*: their transition matrices do not satisfy the Chapman–Kolmogorov composition law except at isolated *division events*, times at which the process momentarily factorizes and behaves Markovianly. Division events are, in this picture, the natural locus of effective measurement and record formation [11, 12].

We adopt the following hypothesis, consistent with this picture and with the gravitational-decoherence proposals [1–3, 19]: when two coherent branches carry distinct effective mass-density records, the gravitational distinguishability of the corresponding geometries acts as a channel that *stabilizes* the branches into distinct records. The stabilization is not a primitive collapse postulate; it is the statement that the gravitational mismatch supplies a positive, accumulating "cost" that suppresses the survival of the unstabilized (coherent) alternative. We make this quantitative in the next two sections and emphasise where the indivisible structure changes the result.

We distinguish three physically separate effects, only the third of which is the subject of this paper:

1. **Environmental decoherence**—ordinary scattering, gas collisions, thermal emission; reducible in principle by better isolation [13, 15].
2. **Unitary gravitational phase**—gravity shifts the relative phase of the branches without destroying coherence.
3. **Gravitational record stabilization**—the geometric mismatch itself acts as a branch-stabilizing (decohering) channel. This is the putative new physics.

III. GRAVITATIONAL MISMATCH ENERGY AND THE MARKOVIAN (DIÓSI–PENROSE) LIMIT

Let the two branches carry mass densities $\rho_r(x)$ and $\rho_{r'}(x)$, with difference $\delta\rho(x) = \rho_r(x) - \rho_{r'}(x)$. Following Diósi and Penrose [1–3], the gravitational mismatch energy is the Newtonian self-energy of the difference,

$$E_G = \frac{G}{2} \int d^3x\, d^3y\, \frac{\delta\rho(x)\,\delta\rho(y)}{|x-y|}. \tag{2}$$

E_G is non-negative once a finite object size is retained; the point-mass limit is not admissible because it diverges, exactly the regularization issue that the experimental bounds constrain [8]. Define the dimensionless gravitational action accumulated over a hold time T,

$$a(T) = \frac{E_G\, T}{\hbar}, \qquad \tau_G = \frac{\hbar}{E_G}. \tag{3}$$

The standard DP result is the exponential coherence law

$$C(T) = C(0)\, \exp(-T/\tau_G). \tag{4}$$

It is instructive to see precisely which assumption produces the exponential. Write the survival factor $S(E_G, T) = |C(T)|/|C(0)|$. If one assumes (i) time multiplicativity $S(E_G, T+U) = S(E_G, T)\, S(E_G, U)$ and (ii) dependence only on the dimensionless action a (additivity of independent cost channels then follows and need not be assumed separately), the Cauchy functional equation forces

$$S = \exp(-\kappa\, a) = \exp(-\kappa\, E_G T/\hbar), \tag{5}$$

with the coefficient κ fixed by normalization. Assumption (i) is the semigroup (Chapman–Kolmogorov) property: it states that processing over the whole interval factorizes into independent processing over its parts. **This is the divisibility / Markov assumption.** (This functional argument is not the historical derivation of DP, which follows from a Markovian master equation [2, 19]; it serves only to isolate the semigroup assumption behind *any* exponential survival law.) The DP exponential is therefore the Markovian special case of a more general survival functional. We now drop assumption (i), as the indivisible structure requires.

IV. INDIVISIBLE DYNAMICS AND NON-EXPONENTIAL DECOHERENCE

A. A minimal finite-memory model

Indivisibility motivates *finite memory*, but a specific coherence law needs an explicit model. We state one as a benchmark and label every assumption, since not all of them come from ISP. Let the relative phase between the branches accumulate from a real gravitational record-mismatch energy $\delta E(t)$,

$$\varphi(T) = \frac{1}{\hbar}\int_0^T \delta E(t)\, dt, \qquad \frac{C(T)}{C(0)} = \langle e^{i\varphi(T)} \rangle, \tag{6}$$

and posit:

- **(M1) Gaussian, stationary noise—standard.** $\delta E(t)$ is a stationary Gaussian process with covariance $K(s) = \mathrm{Cov}[\delta E(t), \delta E(t+s)]$; then the cumulant expansion truncates at second order and the coherence modulus is $|\langle e^{i\varphi} \rangle| = \exp(-\frac{1}{2}\,\mathrm{Var}\,\varphi)$ exactly (any nonzero mean of δE contributes only the unitary phase of Sec. II). This is the ordinary Gaussian-dephasing model, not specific to ISP.
- **(M2) Amplitude set by E_G—the DP ansatz.** The fluctuation scale is the gravitational self-energy of the mass-density difference, $\sigma_E \sim E_G$. Note δE is a genuine *fluctuation*, not a definite phase rate: a definite δE would give only the unitary phase of Sec. II, and it is the *randomness* of δE that decoheres. That the gravitational self-energy of a superposition is *indefinite*—hence a noise of scale E_G rather than a phase—is the Diósi–Penrose hypothesis, taken over unchanged.
- **(M3) Finite memory—the ISP-motivated part.** $K(s)$ has a finite correlation (record-memory) time τ_c and is *not* white. This is the one ingredient indivisibility supplies: a non-divisible process does not decorrelate instantaneously, so the white-noise idealization underlying DP is dropped.

Under (M1) the coherence obeys the standard second-cumulant decoherence functional [14, 20] (we write the decoherence exponent $\chi(T)$, reserving Γ for Barandes' transition matrix)

$$\ln\frac{|C(T)|}{|C(0)|} = -\,\chi(T), \qquad \chi(T) = \frac{1}{2\hbar^2}\int_0^T\!\!\int_0^T K(t_1 - t_2)\, dt_1\, dt_2. \tag{7}$$

It is convenient to name the *noise power* $D_E \equiv \sigma_E^2\, \tau_c$, so the zero-frequency weight is $S(0) = \int K\, ds = 2D_E$. Only (M3) is ISP's; (M1) is standard and (M2) is DP's, and everything below is the consequence of (M3)—that K is not white—for the *shape* of $C(T)$. Two limits bracket the behaviour.

a. Markovian (white-noise) limit ($\tau_c \to 0$ at fixed noise power D_E, so $K \to 2D_E\,\delta$): the double integral is linear in T, $\chi(T) = (D_E/\hbar^2)\,T$, recovering the DP exponential and identifying $1/\tau_G = E_G/\hbar$ when $D_E = E_G\hbar$. This limit fixes the *power*, not the amplitude: as $\tau_c \to 0$, $\sigma_E \to \infty$.

b. Fully-correlated limit ($\tau_c \to \infty$, an idealized limit in which the hold spans a single indivisible transition with no internal decorrelation—not the literal content of indivisibility, which is only the failure of factorization): here it is instead the *amplitude* σ_E that is held fixed, $K \approx \sigma_E^2$ is constant, so

$$\chi(T) = \tfrac{1}{2}\Big(\frac{\sigma_E T}{\hbar}\Big)^2, \tag{8}$$

a quadratic exponent and hence a Gaussian coherence decay.

B. The prediction hierarchy

It helps to separate what finite memory *forces* from what the benchmark closure *adds*.

a. Generic finite-memory onset—closure-independent. Expanding the functional for $T \ll \tau_c$, where $K(t_1 - t_2) \approx \sigma_E^2$, gives a quadratic exponent

$$\boxed{\chi(T) = \frac{\sigma_E^2}{2\hbar^2}\, T^2 + \text{higher order,}} \tag{9}$$

a *zero-initial-slope* decay whatever the kernel's detailed shape. This is the robust, falsifiable content; it does not depend on identifying σ_E with E_G.

b. Benchmark closure. The two scales are fixed by the self-consistent closure (derived in Sec. IV C from the division-event model), $\sigma_E \sim E_G$ and $\tau_c \sim \tau_G$—the unique closure built from the single available scale E_G (its only time is $\hbar/E_G$, its only energy E_G), reinforced by the argument that the record-memory time equals the stabilization time.

c. Benchmark prediction. In the onset region $T \lesssim \tau_c$ the closure gives a Gaussian at DP's scale,

$$\boxed{C(T) = C(0)\, \exp\Big[-\tfrac{1}{2}\,(T/\tau_G)^2\Big], \qquad \tau_G = \hbar/E_G,} \tag{10}$$

crossing over to a DP-slope exponential tail at later times.

For a generic finite τ_c with the exponential memory kernel $K(s) = \sigma_E^2\, e^{-|s|/\tau_c}$ the full closed form is

$$\chi(T) = \frac{\sigma_E^2\, \tau_c^2}{\hbar^2}\Big(\frac{T}{\tau_c} - 1 + e^{-T/\tau_c}\Big), \tag{11}$$

Gaussian for $T \ll \tau_c$ and linear (DP-slope) for $T \gg \tau_c$. The long-time slope is set by the noise power $D_E = \sigma_E^2\, \tau_c$, the onset curvature by the amplitude σ_E^2, so the Gaussian onset time is $\tau_{\text{Gauss}} = \sqrt{\tau_c\, \tau_G}$, equal to τ_G only at the closure $\tau_c \sim \tau_G$. One must therefore *not* recover DP by taking $\tau_c \to 0$ at fixed σ_E, where the slope vanishes; the strict DP exponential is the white-noise limit (D_E held at $E_G\hbar$, $\tau_c \to 0$, $\sigma_E \to \infty$), in which the Gaussian onset window closes. (The divergent amplitude $\sigma_E \to \infty$ of that limit is the source of white-noise DP's ultraviolet radiation problem; a finite τ_c band-limits the noise in time—a *temporal* regulator complementary to DP's *spatial* regularization R_0, see Sec. VIII.) At the closure $D_E = E_G\hbar$ automatically, so the late-time tail carries exactly the DP slope $1/\tau_G$: a single self-consistent law (Gaussian onset, DP-slope tail, crossover at τ_G), not a one-parameter interpolation at fixed amplitude.

The short-time quadratic onset is not an artefact of the exponential kernel; it is the universal consequence of finite-correlation-time (smooth) noise, identical in origin to the quantum-Zeno short-time behaviour of any open system [21], and is required by $\chi'(0) = 0$ whenever the noise has no white (delta) component.

d. What is forced, inherited, and assumed. In sum: the *quadratic onset* (zero initial slope) is forced by finite memory (M3); the *energy scale* E_G is inherited from Diósi–Penrose (M2); the *Gaussian form at* τ_G follows only after the benchmark closure. The model is therefore a finite-memory, DP-family phenomenology *motivated* by ISP—not a first-principles derivation of the memory kernel from indivisible dynamics.

C. A microscopic division-event model

The exponential kernel and the scale τ_c need not be assumed—both follow from an explicit indivisible model of the channel. Model the gravitational record channel as a *renewal-reset* process: $\delta E(t)$ is held constant between Barandes division events and redrawn (the record stabilizing) at each, with division events forming a renewal process of mean spacing τ_c. The stationary autocovariance is then

$$K(s) = \sigma_E^2\, P_{\text{same}}(s), \qquad P_{\text{same}}(s) = \frac{1}{\tau_c}\int_s^\infty (\tau - s)\, \psi(\tau)\, d\tau, \tag{12}$$

fixed by the division-waiting-time law ψ. **Memoryless (Poisson) division events give exactly the exponential kernel** $K(s) = \sigma_E^2 e^{-|s|/\tau_c}$ assumed above, with τ_c the mean inter-division time; regular divisions give instead a triangular kernel—so the kernel shape encodes the division statistics, a microstructure that colored-noise CSL lacks.

The scale τ_c is then fixed self-consistently rather than dimensionally: a division event *is* a record stabilization, which occurs once the channel has accumulated order-unity decoherence, $\chi(\tau_c) \approx 1$. With $\sigma_E \sim E_G$,

$$\chi(\tau_c) = \left(\frac{\tau_c}{\tau_G}\right)^2 e^{-1} = 1 \implies \tau_c \approx 1.65\,\tau_G, \tag{closure}$$

pinning the crossover to τ_G (see Sec. VII for the alternative candidate scales). This converts the two inputs of Secs. IV A–IV B—the exponential kernel and $\tau_c \sim \tau_G$—into consequences of a single assumption: that the gravitational record channel undergoes memoryless division events.

We are explicit that the renewal-reset dynamics is itself a model. It makes Barandes' division events concrete and reproduces the kernel, but deriving it from the full configuration-space indivisible process for a gravitationally-coupled system remains open. So this grounds the kernel one level deeper—in the division-event structure—without claiming a first-principles derivation.

(One bookkeeping caveat on the closure: the renewal-reset process is a jump process, not jointly Gaussian, so the second-cumulant χ used in the closure is exact only to the same $O(10\%)$ non-Gaussian corrections quantified in the third-axis discussion; the fixed point $\tau_c \approx 1.65\,\tau_G$ inherits that accuracy, which is ample for its role as a scale estimate.)

a. On the state-dependence this introduces. Because $\tau_c \approx 1.65\,\tau_G \propto 1/E_G$, the record-memory time scales with the experimentally-chosen mass configuration. This is *not* a dynamical backreaction or a Schrödinger–Newton nonlinearity: $\chi(\tau_c) \approx 1$ is an algebraic self-consistency condition that fixes a parameter once E_G is given, not a self-potential sourced by the wavefunction (the mean-field Schrödinger–Newton effect is a deterministic phase *shift* with no decay—a separate entry in the Sec. V classification, testable by the proposed optomechanical Schrödinger–Newton experiments [23]). For a fixed preparation the dynamics is an ordinary *linear* dephasing channel, and the E_G-dependence of τ_c is the same kind of state-dependence as the E_G-dependence of the decay *rate* $1/\tau_G$ in standard DP/CSL. What differs is the bookkeeping: DP/CSL obtain that dependence as an *emergent* matrix element of a master equation with state-independent coefficients (R_0, λ), whereas here τ_c is fixed *from* the two-branch E_G. The relation $\tau_c \sim \tau_G$ is therefore an **effective, two-branch description**—adequate for the interferometric test, where E_G is a fixed number—and not yet a fundamental, state-independent law for arbitrary (multi-branch, continuous) superpositions; supplying that law is part of the same open problem as grounding the division dynamics in Barandes' formalism.

V. A SHAPE-BASED INTERFEROMETRIC DISCRIMINATOR

The practical signature is the *shape* of the decay near $T = 0$; the discriminating feature is its leading power in T, which is *rate-robust*—it does not rely on an absolute collapse-rate fit. Prepare a spatial superposition with fixed gravitational mismatch E_G and a fixed environmental budget $\Gamma_{\rm env}$, and measure the interferometric visibility $V(T)$. Define the residual gravitational log-coherence by subtracting the separately measured environmental contribution (cleanly, by the density lever below — same size, shape and surface at lower bulk density; the cruder $E_G \to 0$ small-separation control changes the environmental rate too, since $\Gamma_{\rm env}$ itself depends on d below the environment's coherence wavelength, and is usable only with that d-dependence modeled out),

$$L(T) = \ln \frac{V(T)}{V_{\rm env}(T)}. \tag{13}$$

The three hypotheses give qualitatively distinct onsets:

$$L(T) = \begin{cases} 0, & \text{ordinary quantum mechanics (no fundamental channel),} \\ -T/\tau_G, & \text{Markovian collapse: finite initial slope (rate shown for DP; CSL's rate is set by } (\lambda, r_C)), \\ -\frac{1}{2}(T/\tau_G)^2, & \text{indivisible ISP: zero initial slope (parabola tangent to axis).} \end{cases} \tag{14}$$

The discriminator is therefore: a straight line through the origin ⇒ Markovian DP/CSL; a parabola tangent at the origin ⇒ indivisible ISP (Fig. 1). Because the shape (linear vs quadratic onset) is independent of the overall coefficient, the test does not require knowing τ_G in advance, nor the absolute collapse strength—only the ability to vary E_G while holding the environment fixed, and to resolve the curvature of $L(T)$ at small T (distinguishing a small linear slope from a small quadratic curvature is itself a statistical task, but one that needs no absolute-rate calibration).

a. How large is the effect, and what precision does it need? At fixed E_G the two candidate visibility curves are $V_{\rm exp} = e^{-x}$ (Markovian DP) and $V_{\rm Gauss} = e^{-x^2/2}$ (benchmark ISP), with $x = T/\tau_G$ (Table I). The curves separate most near $T \approx 0.65\,\tau_G$, where they differ by $\Delta V \approx 0.29$—about 30% of the initial visibility. Discriminating the two

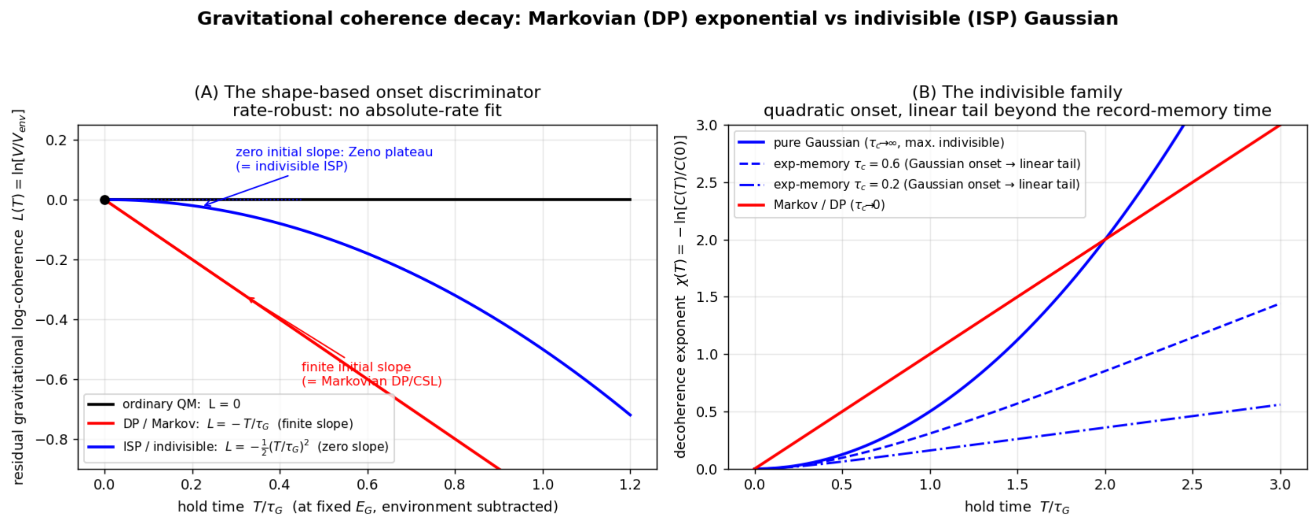

FIG. 1. The shape-based discriminator. **(A)** Residual gravitational log-coherence $L(T) = \ln[V/V_{\rm env}]$ at fixed E_G: ordinary quantum mechanics gives $L = 0$; Markovian DP/CSL a straight line through the origin (finite initial slope); the indivisible finite-memory channel a parabola tangent to the axis (zero initial slope, a Zeno-like plateau). The discriminating feature is the leading power of T, which needs no absolute-rate calibration. **(B)** The decoherence exponent $\chi(T) = -\ln[C(T)/C(0)]$ for the finite-memory family: a pure Gaussian ($\tau_c \to \infty$), exponential-memory kernels with finite τ_c (Gaussian onset crossing to a linear, DP-slope tail beyond the record-memory time τ_c), and the Markovian DP limit ($\tau_c \to 0$). The benchmark closure sits at $\tau_c \sim \tau_G$.

shapes at the few-σ level therefore calls for a visibility resolution of a few percent across the window $T \in [0.3, 1.5]\,\tau_G$ (with τ_G of order seconds to hours for the masses of Sec. VI)—demanding, but on the trajectory of current mesoscopic-superposition programmes. In the *defined* observable $L = \ln(V/V_{\rm env})$ the separation is exactly $\Delta L = x - x^2/2$, which peaks at $x = 1$ (i.e. $T = \tau_G$, $\Delta L = 0.5$); since interferometric data are usually analysed in log-visibility, $T \approx \tau_G$ is the natural probe time.

TABLE I. Candidate visibility curves and their separation, $x = T/\tau_G$.

$x = T/\tau_G$	$V_{\rm exp}$	$V_{\rm Gauss}$	ΔV
0.3	0.74	0.96	0.21
0.5	0.61	0.88	0.28
0.65	0.52	0.81	0.29
1.0	0.37	0.61	0.24
1.5	0.22	0.32	0.10

A non-Markovian *environment* produces the same quadratic onset [20, 21], so curvature alone does not establish a gravitational origin—only the *scaling* with E_G does, and the discriminating power lies in knobs that move E_G while leaving the environment fixed:

- **Density.** Gravitational decoherence scales as $E_G \propto \rho^2$ (harmonic $\propto \rho^2 R^3 d^2$, saturated $\propto \rho^2 R^5$), whereas gas-collision and blackbody decoherence depend on external geometry, surface, polarizability, pressure and temperature but are essentially *blind to bulk mass density* at fixed external size [15] (with one honest caveat: dielectric response correlates with density across real materials via Clausius–Mossotti, so the blackbody channel's ρ-blindness is approximate, and its discrimination leans on the internal-temperature knob below). Comparing same-size, same-shape particles of very different density isolates gravity by a ρ^2 law no environmental channel mimics.

- **Common-mode subtraction.** Run a high-E_G and a low-E_G configuration in the *same* chamber, vacuum and thermal/seismic environment and subtract: vibration, gas and blackbody backgrounds are largely common-mode and cancel, leaving the gravitational differential—the standard precision-measurement strategy.

- **Calibrate by knob.** Gas decoherence scales with pressure and blackbody with internal temperature, while gravity is pressure- and temperature-independent; mapping the onset curvature versus P and T and extrapolating $P \to 0$, $T \to 0$ removes those channels.

Two further fingerprints sharpen this. (a) Gravitational $E_G(d)$ saturates at the *object size* $d \sim R$, whereas environmental channels saturate at thermal or scattering wavelengths. (b) **Plateau duration.** The quadratic (Zeno) plateau of *any* finite-memory channel lasts only until its correlation time; the gravitational plateau persists to $\tau_c \sim \tau_G$ (seconds–hours), whereas gas- and blackbody-bath correlation times are microseconds or shorter—so a quadratic onset *surviving to macroscopic hold times* cannot be a fast environmental bath, which would already have crossed over to its linear tail. This cleanly excludes the *fast* baths; it does **not** by itself exclude slow, low-frequency backgrounds (structural $1/f$ noise, drift), which can sustain a long plateau of their own—those are removed not by plateau duration but by the E_G-scaling (density) lever above, which no environmental channel reproduces. None of this is easy—it is a long-term programme, not a near-term measurement—but the gravitational channel is identified by a *joint* scaling (with ρ, d, P, T) that no single environmental source reproduces.

b. A second axis: distinguishing ISP from the whole collapse-model family. The onset shape separates ISP from any *Markovian* collapse model, but several related theories must be told apart at once, and they are distinguished by *two* axes—the onset shape and the scaling of the rate. (Both axes are plain, non-relativistic physics: indivisibility supplies the onset, and the *inherited* DP scale — ansatz (M2), not an ISP derivation — supplies the E_G scaling; the relativistic extension adds nothing here.) This cell—finite-memory gravitational decoherence—is reached by no Markovian or non-gravitational model (Table II).

TABLE II. The two-axis classification: onset shape × rate scaling.

Theory	Fundamental decay?	Onset shape	Rate scaling
Unitary QM / QFT	no	— (only $\Gamma_{\rm env}$)	—
Schrödinger–Newton (mean-field) [23]	no—a phase/freq. *shift*	—	grav., but a shift
CSL / GRW (Markovian, non-grav.) [4, 5]	yes	exponential	params (λ, r_C), not E_G
Diósi–Penrose (Markovian, grav.) [1, 2]	yes	exponential	gravitational E_G
ISP finite-memory (this work)	**yes**	**Gaussian**	**gravitational** E_G

A two-axis measurement settles it. **Axis 1 (onset shape)**, above, separates ISP from the Markovian models (exponential vs Gaussian). **Axis 2 (rate scaling):** vary the mass M, separation d, and geometry, and test whether the rate tracks the gravitational self-energy E_G (the Diósi–Penrose form, $\propto G$) or the CSL parameters $(\lambda, r_C$, with a fixed $\sim 10^{-7}$ m length scale and a different geometry dependence); the Schrödinger–Newton alternative appears instead as a deterministic frequency *shift* with no decay, testable by the proposed optomechanical experiments [23]. A result in the (Gaussian onset) × (gravitational E_G-scaling) cell supports the ISP-motivated non-Markovian gravitational hypothesis—without uniquely establishing ISP (see the limits below)—and simultaneously excludes Markovian DP (exponential), CSL/GRW (wrong scaling), Schrödinger–Newton (shift, not decay), and unitary QM (no decay).

c. A third axis: the non-Gaussian fingerprint of discrete records. A *coloured* (non-Markovian) *gravitational* collapse—Gaussian noise with the same E_G-scaling—would share both cells above. A third axis separates it, and it too lives in the visibility curve. Generic coloured noise is Gaussian by construction, so its coherence is exactly the second-cumulant result $C(T) = e^{-\chi(T)}$ (monotone, log-convex); the division-event process of Sec. IV C is a *jump* process and is non-Gaussian,

$$\ln |C(T)| = -\chi(T) + \frac{\kappa_4(T)}{24} - \cdots , \tag{15}$$

where the Gaussian alternative has all cumulants $\kappa_n = 0$ for $n \geq 3$. In the long-memory regime $T \lesssim \tau_c$ the coherence is the *characteristic function* of the per-division mismatch-energy distribution, $C(T) = \langle e^{i\delta E\, T/\hbar} \rangle$: a Gaussian $p(\delta E)$ reproduces the Gaussian onset, while a discrete, branch-selecting δE produces non-Gaussian structure—up to coherence *revivals* ($C(T) = \cos(\sigma_E T/\hbar)$ for $\delta E = \pm\sigma_E$), which no monotone $e^{-\chi}$ can produce. The effect is not parametrically small: at $T \sim \tau_G$ the leading correction is set by the excess kurtosis of $p(\delta E)$ (for bimodal $\delta E = \pm\sigma_E$ the quartic term is $x^4/12$ against the quadratic $x^2/2$: 17% of the onset exponent at $x = \sigma_E T/\hbar = 1$, equivalently $\approx 8\%$ of $\chi \approx 1$ at the closure point — both conventions stated since an earlier draft quoted only the second), and it occupies the *same* observable window as the onset, washing out only in the white-noise limit. Since the discreteness of records is precisely the ISP content a continuum (Gaussian) theory lacks, a measured departure of $C(T)$ from the pure form $e^{-\chi(T)}$—a T^4 term, or non-monotonicity—is a *positive* signature of discrete record formation. The three axes nest: (1) onset shape (linear vs quadratic) falsifies strict Markovian DP/CSL; (2) rate scaling (E_G vs λ, r_C) separates gravitational from non-gravitational; (3) non-Gaussianity (a T^4 term, up to revivals) separates discrete-record from Gaussian coloured gravitational collapse. Axis 3 rides on $p(\delta E)$ being non-Gaussian— unpredicted from first principles (a limitation, below)—and narrows the field to *discrete-record* models rather than singling out ISP uniquely.

d. Honest limits of the test. (i) A non-Markovian *gravitational* collapse model could share the first two cells; axis 3 separates the Gaussian (coloured-noise) version by its absent non-Gaussian fingerprint, narrowing the degeneracy to *discrete-record* gravitational models, which are distinguished from ISP only at the level of motivation and ontology. (A *coloured* (non-Markovian) CSL [10] can mimic the Gaussian onset but not the gravitational E_G-scaling, so the second axis still separates that case.) (ii) Axis 2 is a *campaign*: the E_G-scaling must be mapped across a range of M, d, geometry, not read from one curve. (iii) The test presupposes the fundamental channel exists at all—if nature is exactly unitary, every row collapses to "no decay" and ISP is empirically QM. (iv) The regime is at the frontier of mesoscopic-superposition sensitivity, not yet in hand. Within these limits the test upgrades the discriminator from "ISP vs. standard quantum mechanics" to "ISP vs. the entire collapse/decoherence family."

VI. NUMERICAL THRESHOLDS AND EXPERIMENTAL PLATFORMS

We quote thresholds for a rigid homogeneous sphere of mass M, radius R, density ρ, coherently separated by d. For $d \ll R$ the mismatch energy is harmonic,

$$E_G \simeq \tfrac{1}{2} M \omega_G^2 d^2, \qquad \omega_G^2 = \frac{4\pi G \rho}{3}, \tag{16}$$

while for object-scale separations it saturates at the self-energy scale

$$E_G^{\mathrm{sat}} \simeq \frac{6}{5} \frac{GM^2}{R}. \tag{17}$$

Using $\rho = 2000\,\mathrm{kg\,m^{-3}}$, the saturated threshold mass that decoheres in time T (i.e. $\tau_G = T$) is

$$M_{\mathrm{thr}}^{\mathrm{sat}}(T) = \left[\frac{\hbar}{\frac{6}{5} G\,(4\pi\rho/3)^{1/3}\, T} \right]^{3/5}. \tag{18}$$

Representative values are given in Table III (saturated regime, $d \gtrsim R$; order-of-magnitude DP-energy thresholds at $\tau_G = T$, not feasibility estimates). For an actual experiment the self-energy must be evaluated numerically from the measured density profile and branch geometry; the harmonic and saturated formulas above are limiting estimates used only to locate the relevant mass–time scale. These mass and separation windows coincide with the targets of current and proposed mesoscopic-superposition experiments [15, 16, 18, 22, 25–27]. Three platform classes are natural hosts for the onset-shape test:

- **Levitated nanosphere interferometry.** Neutral dielectric nanospheres in ultra-high vacuum, prepared in spatial superposition and recombined; ground-state cooling and control of such particles is now routine [26, 27].
- **Mesoscopic matter-wave interferometry.** Large-molecule and cluster interferometers, where macroscopicity has been advanced systematically, now beyond 25 kDa [15, 22, 25].
- **Mechanical cat states / optomechanics.** Superpositions of a massive mechanical element [18], including the gravity-mediated-entanglement geometries [16, 17] in which E_G can be tuned by separation.

In every case the experimental strategy for the discriminator is identical: hold the object, temperature, vacuum, and pulse sequence fixed, vary E_G (e.g. through the separation d), and examine whether the gravitational contribution to $\ln V$ enters linearly or quadratically in the hold time.

VII. THE RECORD-MEMORY TIME

The new scale τ_c is the autocorrelation time of the gravitational mismatch—in the ISP language, the mean spacing between division events of the gravitational record channel. Section IV C fixed it for the *channel-intrinsic* case—where the division event *is* the record stabilization—at $\tau_c \approx 1.65\,\tau_G$ via the record-stabilization fixed point $\chi(\tau_c) \approx 1$, so that the memory time and the stabilization time coincide up to an $O(1)$ factor,

$$\tau_c \sim \tau_G = \hbar / E_G. \tag{19}$$

TABLE III. Saturated threshold mass and radius decohering in time T ($\rho = 2000\,\mathrm{kg\,m^{-3}}$).

T	$M_{\rm thr}$ (kg)	$M_{\rm thr}$ (amu)	$R_{\rm thr}$
$1\,\mu$s	3.1×10^{-12}	1.9×10^{15}	$7.2\,\mu$m
1 ms	4.9×10^{-14}	2.9×10^{13}	$1.8\,\mu$m
1 s	7.7×10^{-16}	4.6×10^{11}	452 nm
100 s	4.9×10^{-17}	2.9×10^{10}	180 nm
1 h	5.7×10^{-18}	3.4×10^{9}	88 nm

That value is not unique, however: it presumes the channel is governed by its own stabilization time, whereas other physical scales are conceivable. The candidate scales and their consequences are given in Table IV.

Two considerations show the channel does not generically slide to the Markovian floor, and that $\tau_c \gtrsim \tau_G$ is *strongly motivated* rather than freely chosen (not forced: the information-gating premise below — that record formation requires order-unity distinguishability — is itself a hypothesis). First, **records are information-gated**: a division event is the gravitational field committing to a which-branch record, and no record can form before the branches are distinguishable—at the light-crossing time the accumulated distinguishability is $\chi(R/c) \sim (R/c\,\tau_G)^2 \sim 10^{-30}$, so there is nothing to record, and the record time is bounded below by the time to reach order-unity distinguishability, $\tau_c \gtrsim \tau_G$. Second, **a fast clock does not recover DP** at the physical fluctuation amplitude $\sigma_E \sim E_G$: the decoherence slope $\sigma_E^2 \tau_c/\hbar^2$ is then smaller than the DP slope $1/\tau_G$ by $\tau_c/\tau_G \sim 10^{-15}$, a quantum-Zeno freeze, not the DP exponential. (Recovering DP as $\tau_c \to 0$ requires holding the noise *power* $\sigma_E^2 \tau_c$ fixed, hence $\sigma_E \to \infty$; see Sec. IX.) Equivalently, requiring the decoherence to be DP-*sized* at $\sigma_E \sim E_G$ forces $\tau_c \gtrsim \tau_G$: the onset curvature reaches order unity at $T \sim \tau_G$ only if the crossover has not already turned the decay into its slow linear tail.

The honest verdict is that *finite memory* forces the decay away from a strict exponential—and indivisibility motivates the finite memory—with τ_c tied to the gravitational scale two ways: by the Sec. IV C fixed point and by information-gating. (A third consideration sometimes invoked—that an *observable* DP-scale effect requires τ_c in the measurement window—is experimental-design reasoning, not physics: nature is not obliged to put τ_c in the observable window, and we do not count it as a pinning.) The DP-Markovian limit is the unphysical $\sigma_E \to \infty$ corner. The channel-intrinsic and gravitational-dynamical candidates both place the crossover within the observable window (τ_c of order seconds to hours for the masses in Sec. VI), so the Gaussian onset is accessible.

TABLE IV. Candidate record-memory times and their observational consequence.

candidate τ_c	origin	consequence
$\hbar/E_G = \tau_G$	channel-intrinsic (division = stabilization)	Gaussian onset near $T \sim \tau_G$
$\sqrt{3/4\pi G\rho} = 1/\omega_G$	gravitational dynamical time (density only)	$\sim 10^3$ s for rock: observable
R/c	light-crossing (microphysical floor)	$\sim 10^{-15}$ s: Zeno-suppressed at fixed σ_E ($\sigma_E \to \infty$ for DP)

VIII. SPONTANEOUS RADIATION AND EXISTING BOUNDS

The parameter-free DP model is experimentally excluded. Donadi et al. [8] searched underground at the Gran Sasso laboratory for the spontaneous electromagnetic radiation that collapse-driven charge diffusion would produce, found no excess, and tightened the bound on the effective nuclear mass-density size by about three orders of magnitude, ruling out the simplest DP model; current bounds on the wider collapse-model family are reviewed in [24].

The radiation bound constrains the model's regularization, not its prediction. The DP noise carries a short-distance regularization length R_0, and two effects are set by *different* scales:

- the **decoherence** of the superposition is collective—the gravitational self-energy of the displaced mass distribution, $E_G \approx (6/5)GM^2/R$, set by the *object size* R and **independent of** R_0 (every nucleon separation exceeds R_0);
- the **spontaneous emission and heating** of bulk matter are a per-nucleon momentum diffusion, $\propto 1/R_0^3$, set by the regularization R_0.

These decouple: for any nanoscale object the structural (decoherence) term exceeds the per-nucleon term by many orders, so matching the decoherence does *not* fix the emission. The Gran Sasso X-ray search therefore bounds R_0—

giving $R_0 \gtrsim 0.5$ Å for white DP [8], with spontaneous-heating and mechanical bounds two orders weaker ($R_0 \gtrsim 5 \times 10^{-13}$ m) but robust to non-Markovian generalizations [24]—and none of these touch the (R_0-independent) decoherence the interferometer measures.

The finite-memory modification only *helps* on the emission side. Emission at the photon frequency is governed by the noise spectral density there—a noise band-limited to $1/\tau_c$ is adiabatic on the keV photon timescale and carries no keV quanta to deposit—so with $S(\omega) = S(0)/[1 + (\omega\tau_c)^2]$ the temporal cutoff suppresses keV emission by $(E_G/\hbar\omega_{\mathrm{keV}})^2 \sim 10^{-36}$ relative to white DP, the gravitational mismatch energy being the natural emission cutoff. This Lorentzian (rather than exponential) tail is the *least*-suppressed case, forced by the cusp the discrete division events put in the kernel at the origin ($K'(0^+) = -\sigma_E^2/\tau_c$, for any waiting-time law); even this worst case is suppressed by 10^{-36}; with sharp lab-time resets the emission margin would read out the division microstructure of Sec. IV C, though a covariant completion smooths the kernel's cusp on timescales $\lesssim R/c$ and weakens that readout while strengthening the suppression (see the Note added). In this sense the finite memory acts as a *temporal* ultraviolet regulator: the band-limit $1/\tau_c$ curtails the high-frequency radiation that the flat white-noise spectrum would make divergent—the role the *spatial* smearing R_0 plays for the static self-energy. The two regulators act on *different* sectors and do not substitute for one another: τ_c does not soften the (R_0-set) spatial self-energy, nor the (zero-frequency, $S(0)$-set) heating, which is precisely why the heating bound remains the robust one above. (The zero-frequency emission term sometimes invoked for collapse models was shown to be a non-normalizable-state artifact that vanishes for wave packets [9, 10].) The finite-memory channel thus inherits standard DP's consistency with the radiation and heating bounds—at worst identically, and at the keV frequencies actually searched, more comfortably—while changing only the *onset shape* of the decoherence, not its magnitude or radiation phenomenology. (One extrapolation is used here and flagged: the bulk-matter emitters of the radiation searches carry no prepared superposition, while τ_c in this model is tied to the two-branch configuration of an interferometer — the effective-description caveat of Sec. IV C. The conclusion is insensitive to this: *any* τ_c far above attoseconds kills keV emission, and the zero-frequency heating bound is inherited at worst identically.)

IX. DISCUSSION: SCOPE AND LIMITATIONS

We state the boundaries of the result plainly.

- **The energy scale is DP's, not new.** The contribution is the non-exponential *shape* and its falsifiable onset, not the magnitude $\tau_G = \hbar/E_G$, which is taken over from [1–3].
- **The coefficient inherits DP's ansatz dependence.** Whether the Gaussian onset time equals $\hbar/E_G$ exactly depends on τ_c and σ_E (generally $\tau_{\mathrm{Gauss}} = \sqrt{\tau_c\,\tau_G}$); only the quadratic *shape* is assumption-light.
- **The quadratic onset is generic to non-Markovianity.** It is shared by coloured-noise CSL and by non-Markovian environmental decoherence [10, 20, 21]. Observing it would falsify *strict* Markovian DP/CSL and be evidence for a non-Markovian fundamental channel consistent with ISP, but would not by itself single out ISP among non-Markovian models.
- τ_c **is a genuinely new parameter.** The division-event model (Sec. IV C) pins it to $\approx 1.65\,\tau_G$ via the record-stabilization fixed point, but this rests on the renewal-reset ansatz, and Sec. VII lists other candidate scales that interpolate toward the Markovian limit.
- **The analysis is non-relativistic throughout.** Kernels are correlated in laboratory time, energies are Newtonian, and the platforms are slow: nothing here claims Lorentz covariance, and the covariant completion of a finite-memory kernel is a separate problem (see the Note added).
- **The DP limit is the white-noise limit, not $\tau_c \to 0$ at fixed amplitude.** At fixed σ_E the $\tau_c \to 0$ limit gives vanishing decoherence, not DP; the strict DP law requires holding the noise power $\sigma_E^2\,\tau_c$ fixed. The clean "Gaussian onset at τ_G with a DP-slope tail" picture holds at the self-consistent point $\tau_c \sim \tau_G$, $\sigma_E \sim E_G$ (Secs. IV B, IV C and VII), which is an ansatz, not a theorem.
- **Environmental non-Markovianity must be controlled.** Because a non-Markovian *environment* can also produce a quadratic onset, the test requires that the residual after environmental subtraction be attributable to the gravitational channel—most cleanly demonstrated by its scaling with E_G at fixed environment.

Within these bounds, the result is a concrete, falsifiable alternative to the Markovian DP law, motivated by a specific and independently-motivated stochastic ontology [11, 12], and testable with the same platforms already pursuing DP [15, 16, 18, 22].

X. CONCLUSION

Treating gravitational decoherence as a process in an indivisible (non-Markovian) stochastic dynamics replaces the Diósi–Penrose exponential with a non-exponential, Gaussian-onset coherence decay—at the self-consistent point $\tau_c \sim \tau_G$, $\sigma_E \sim E_G$, at the same gravitational energy scale—with the strict DP exponential recovered in the white-noise limit. The distinction is experimentally sharp and rate-robust: the discriminator is the leading power of T in the residual gravitational log-coherence—linear for Markovian collapse, quadratic for the indivisible channel—not an absolute rate. The benchmark model identifies the crossover scale τ_c with the DP time τ_G—other choices interpolate between a visibly non-Markovian onset and the Markovian DP limit—and with $\tau_c \sim \tau_G$ the effect lies within reach of mesoscopic-superposition experiments. The robust content is conceptual and methodological—that the DP exponential is a Markovian idealization, and that the onset *shape* together with its E_G-scaling is a falsifiable discriminator for *any* finite-memory gravitational decoherence, separable by a density lever and common-mode subtraction from environmental backgrounds. The radiation and heating bounds constrain only the model's regularization length and decouple from the (R_0-independent) decoherence, so the finite-memory channel inherits standard DP's consistency with them; what it adds to standard DP is the onset *shape* alone—now grounded (Sec. IV C) in the division-event structure of the indivisible process.

NOTE ADDED (11 JUNE 2026)

After this paper was completed, companion work within the same program (manuscript in preparation: *Interacting Tomonaga–Schwinger integrability of indivisible division-event dynamics*; archived with the program repository accompanying this deposit) examined the *relativistic* completion of finite-memory collapse noise and reached two results that bear on, without modifying, the present analysis. (i) A frame-fixed correlation time — the form used here, $K(s) \propto e^{-|s|/\tau_c}$ in laboratory time — is not Lorentz-covariant; a covariant memory must correlate in the invariant interval, $K = f\big((x-y)^2\big)$, and finiteness of the induced energy production requires its spectrum to decay faster than any power (a quartic-exponential class $\hat{f}(q) \propto e^{-q^4/\beta^4}$ suffices), while Lorentzian/Ornstein–Uhlenbeck spectra remain ultraviolet-divergent relativistically. (ii) This does not alter the present paper's results, whose scope is explicitly non-relativistic (Sec. IX), with one qualification: a covariant $f\big((x-y)^2\big)$ restricted to a laboratory worldline is a smooth even function of the time lag, so it reproduces the exponential kernel used here only up to a covariant smoothing of the kernel's cusp at the origin on timescales $\lesssim R/c$ — which leaves the onset-shape discriminator untouched (it is a property of the finite-memory *class*, not of the kernel's detailed shape) and only *strengthens* the high-frequency emission suppression of Sec. VIII, while weakening that section's microstructure-readout remark, which we have qualified accordingly. What the companion result adds is a sharpening at the program level: if the quadratic onset is observed, the covariant completion of the inferred memory is constrained to the smooth (faster-than-power-law) spectral class, not merely to "coloured noise."

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